

Delivery 3 In-editor visualization

The aim of this delivery is for you to create a data visualization toolkit inside Unity that allows gameplay events to be inspected directly in the editor. You will demonstrate your ability to transform raw telemetry into meaningful spatial visualizations such as heatmaps, movement paths, or event distributions, while also providing ways to interact with and filter the data. Your project should show how gameplay data can be collected, stored, and re-imported, but the main focus is on building intuitive and interactive editor tools that give designers actionable insights.

You may complete the assignment in Unity or in any other game engine of your choice. You can work with a project you have already created or use the provided sample Project:

<https://assetstore.unity.com/packages/templates/tutorials/3d-game-kit-lite-135162>

1. Design a visualization strategy

- Decide what gameplay data is worth visualizing spatially.
- Plan how heatmaps, movement paths, or event markers will be displayed in the scene view.

2. Import / Export pipeline

- Export the gameplay data to a MySQL database (with PHP scripts).
- Re-import data back into Unity for visualization.
- Keep the pipeline clean and modular, but the **focus is on visualization, not storage**.

3. Build Unity Editor Visualizations

- Develop custom inspectors or editor windows to display collected data.
- Possible visualizations:
 - **Heatmaps** (e.g., player deaths, item pickups, time spent in areas).

- **Paths** (movement trajectories, patrol routes, player exploration).
 - **Event markers** (icons for specific interactions).
- Add **interaction controls** such as:
 - Grid size adjustments.
 - Color palette or intensity scaling.
 - Filtering by event type or time window.

Deliverables

1. Unity Project

- Scripts for exporting / importing and visualizing data inside the Unity editor.
- A dedicated scene for data visualization with working examples.

2. PHP Scripts & Database Schema

- Minimal set of scripts/tables needed to support your pipeline.

Evaluation criteria

Gather & Store (30%)

Focus: Data structure, sampling strategy, and backend pipeline.

- **9–10 (Excellent)**
 - Data structure is well-designed, scalable, and normalized.
 - Sampling strategy is justified (what, when, and how often data is collected).
 - Event tracking is integrated non-intrusively without altering core gameplay code.
 - PHP and database schema are robust, efficient, and documented.
- **7–9 (Good)**
 - Data structure covers most needs but lacks scalability or flexibility.
 - Sampling strategy works but is not optimized or not well justified.
 - Event tracking works but introduces minor coupling with game logic.
 - Backend is functional but has minor inefficiencies.
- **5–7 (Fair)**
 - Data structure is simplistic or partially incomplete.
 - Sampling strategy is ad hoc, with little explanation.
 - Event tracking works but is intrusive or fragile.
 - Backend is working but poorly structured or hard to maintain.
- **0–4 (Poor)**
 - Data structure is inconsistent or incomplete.
 - No clear sampling strategy.

- Event tracking unreliable or missing.
 - Backend pipeline missing or unusable.
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2. Analytics & Visualization (50%)

Focus: Import/export pipeline, re-integration into Unity, and visualization quality.

- **9–10 (Excellent)**

- Data can be exported, stored, and re-imported seamlessly.
- Visualizations are interactive, customizable, and insightful (e.g., heatmaps with grid control, path overlays, filters).
- Visual tools integrate smoothly into the Unity editor and are easy to use.

- **7–9 (Good)**

- Data can be imported/exported with minor issues.
- Visualizations are functional but limited (basic heatmaps or paths, few interaction options).
- Editor integration works but could be more user-friendly.

- **5–7 (Fair)**

- Import/export is unstable or requires manual intervention.
- Visualizations are minimal or unclear (e.g., only static plots).
- Interaction is limited or missing.
- Unity editor integration feels clunky or incomplete.

- **0–4 (Poor)**

- No working import/export pipeline.
 - Visualizations missing or meaningless.
 - No interaction or Unity integration.
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3. Reporting & Presentation (20%)

Focus: Oral presentation, clarity, and communication of results.

- **9–10 (Excellent)**
 - Oral presentation is clear, concise, and well-structured (within 10 min).
 - Students demonstrate deep understanding of their pipeline, choices, and results.
 - Results are shown with effective visuals and explained in context of gameplay.
 - Communication is professional and engaging.
- **7–9 (Good)**
 - Presentation is clear but could be more structured or polished.
 - Students explain their work but with limited depth or justification.
 - Visuals are used but not optimally explained.
- **5–7 (Fair)**
 - Presentation lacks structure or clarity.
 - Students show partial understanding of their system.
 - Results are shown but not interpreted well.
- **0–4 (Poor)**
 - Presentation is missing, incomplete, or incoherent.
 - Students cannot explain their system or results.
 - No visuals or explanations provided.